

Drought monitoring using an Integrated Drought Condition Index (IDCI) derived from multi-sensor remote sensing data

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Abstract Drought is a complex natural phenomenon. To effectively characterize the spatial extent and intensity of the phenomenon, multiple drought-related factors, such as precipitation, vegetation growth condition, and land surface temperature, should be considered comprehensively. However, the capability of each of these factors in drought monitoring varies with seasonal time. Thus, in formulating a drought index, different weights should be assigned to these factors at different time periods. This study proposes a novel remote sensing index, the Integrated Drought Condition Index (IDCI), for short-term drought monitoring. The index sets different weights for each month of the growing season into three components, i.e., precipitation, vegetation growth condition, and land surface temperature, based on the principle component analysis. To assess IDCI performance, the spatial drought conditions of the IDCI maps during the growing season in a typical dry year and individual month of August from 2003 to 2012 were compared with in situ drought indices in Northern China. Correlation analyses were performed between the IDCI and different timescale Standardized Precipitation Evapotranspiration Index values, and the year-to-year IDCI variations were compared with in situ drought indices. The results of the comparison and correlation analysis confirmed the effectiveness of IDCI in characterizing drought conditions and patterns.

Keywords Drought · MODIS · TRMM precipitation · Principle component analysis · SPEI · Drought index · Drought index weighting

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1 Introduction

Drought is a complex natural hazard. A result of climatic fluctuations and variations, drought triggers negative economic, social, and environmental impacts (Tadesse et al. 2004). In 1995, the Federal Emergency Management Agency (FEMA) estimated that drought affected more people than any other weather-related disasters and the average annual damage in the USA has been estimated to be worth 6–8 billion dollars (FEMA 1995). Drought occurs frequently in China, resulting in considerable impacts and economic losses. The droughts in China have affected an average of 21 million ha annually over the last 50 years, or approximately 20 % of the total cropland area (Jiang et al. 2013). The State Flood Control and Drought Relief Headquarters (SFCDRH) and the Ministry of Water Resources of the Government of the People's Republic of China have reported that droughts were responsible for direct economic losses amounting to RMB 120.6 billion in 2009 (SFCDRH 2011).

Droughts are often classified into four general types: meteorological or climatological, agricultural, hydrological, and socioeconomic (American Meteorological Society 2013). Meteorological or climatological drought is an abnormal phenomenon of water shortages caused by precipitation deficit, which normally triggers the other three drought types (Zhang and Jia 2013). Recent observations have shown that climate change brought about by global warming causes the intensification and increased frequency of droughts, which result in escalated impacts (Dai 2011, 2013; Kogan et al. 2013). Therefore, a comprehensive study on drought monitoring and prediction is important to provide policymakers with accurate and timely information needed for early warning and mitigation.

Drought monitoring worldwide has significantly progressed through various drought indices, such as the Palmer Drought Severity Index (PDSI) (Palmer 1965), Standardized Precipitation Index (SPI) (McKee et al. 1993), Standardized Precipitation Evapotranspiration Index (SPEI) (Vicente-Serrano et al. 2010a), and percentage of precipitation anomalies (Pa) (AQSIQ and SAC 2006; Zhou et al. 2012). These indices are typically calculated using station-based climate and meteorological data. The PDSI is established according to a simplified water balance model that considers long-term historical precipitation, mean temperature data, and the available water content information of soil. The calculation of the SPI is based on statistical probability derived from precipitation data. The main characteristic of the SPI is the flexibility in precipitation anomaly quantification for a specific time period (e.g., 1, 3, 6, 9, or 12 months) according to the long-term precipitation record for specific drought types (Guttman 1998, 1999). Mathematically, SPI values are based only on precipitation data, and it does not consider other variables that can influence droughts. The SPEI is an improved drought index, and it is based on both precipitation and potential evapotranspiration (PET) data. The procedure for calculating the SPEI is similar to that for the SPI, but SPEI considers the role of temperature and uses the difference between precipitation and PET as the input data (Begueria et al. 2014; Stagge et al. 2015; Vicente-Serrano et al. 2010a, b). The SPEI has been widely used in diverse studies because it has great promise as a drought index (Li et al. 2013; Sohn et al. 2013; Wang et al. 2015; Yu et al. 2014). SPEI considers the effect of reference evapotranspiration on drought severity and retains the simplicity multi-scalar nature of the SPI which allows for the identification of different drought types. The Pa is another simple indicator that measures precipitation anomaly compared with historical average for a specific period (Zhou et al. 2012).

The use of station-based indices in the quantitative characterization of drought conditions around meteorological stations in different regions is advantageous. However, in areas with sparse weather stations or in places where available precipitation data are inconclusive, the spatial drought conditions may be estimated using spatial interpolation techniques (Rhee et al. 2010). As a result, detailed characterization and monitoring of the spatial pattern of drought conditions using traditional station-based tools depend on the density and spatial distribution of the stations. This situation highlights the capability of satellite-based remote sensing to quickly obtain a wide range of spatial details and evaluate the relevant information of the drought condition with continuous and accurate data that timely and spatial across large-scale areas (Keshavarz et al. 2014; Malingreau 1986; Son et al. 2012).

The Moderate Resolution Imaging Spectroradiometer (MODIS) data are widely applied in establishing remote sensing drought index, owing to the rich spectral information (Caccamo et al. 2011; Gu et al. 2007; Park et al. 2004; Shahabfar et al. 2012; Son et al. 2012). The Normalized Difference Vegetation Index (NDVI) (Rouse et al. 1974) is widely used in estimating vegetation growth conditions and monitoring drought conditions (Boschetti et al. 2013; Carlson and Ripley 1997; Chen et al. 2004; Hmimina et al. 2013; Kogan 1990; Peters et al. 2002; Tadesse et al. 2005; Xiong et al. 2010). The NDVI fully utilizes the reflective and absorptive characteristics of plants in the near-infrared (NIR) and red channels of the electromagnetic spectrum, which can be obtained as $(\text{NIR} - \text{RED})/(\text{NIR} + \text{RED})$. Kogan (1995) computed the Vegetation Condition Index (VCI) by linearly scaling the NDVI values from 0 to 1 for each pixel according to the absolute maximum and minimum values of the same period in the entire record. Compared with the NDVI, the VCI can measure the impact of weather on vegetation better (Gebrehiwot et al. 2011; Liu and Kogan 1996; Rhee et al. 2010). Surface temperature is also affected by drought, and as such, the Temperature Condition Index (TCI) was developed based on land surface temperature (LST). TCI is widely used in drought monitoring (Bayarjargal et al. 2006; Bhuiyan et al. 2006; Du et al. 2012; Ezzine et al. 2014; Kogan et al. 2012; Seiler et al. 2000). In addition, microwave remote sensing can be utilized in drought monitoring because this method works in all-weather working conditions (Du et al. 2012; Rhee et al. 2010). The Tropical Rainfall Measuring Mission (TRMM) was designed to measure precipitation at a $0.25^\circ \times 0.25^\circ$ resolution from space using various sensors, such as the high-resolution radar, visible-infrared radiometer, and passive microwave radiometer. The precipitation products are widely used to monitor drought conditions effectively (Du et al. 2012; Ezzine et al. 2014; Feng et al. 2012; Keshavarz et al. 2014; Rhee et al. 2010; Zhang and Jia 2013).

Considering the complexity of drought, multiple sources of drought-related factors such as precipitation, vegetation growth condition, and land surface temperature should be integrated to characterize the spatial extent and intensity of drought. Moreover, these factors share correlated information. All these aforementioned points render the principal component analysis (PCA) as capable of providing unique advantages in establishing a remote sensing drought index that considers precipitation, vegetation, and land surface temperature (Du et al. 2012). PCA is an orthogonal linear transformation that converts a set of observations of possibly correlated variables into a new coordinate system that can be represented without correlation (Baert et al. 2012; Lasaponara 2006; Liu et al. 2014; Vicente-Serrano et al. 2004; Xia et al. 2014). PCA is widely used in analyzing spatial drought patterns (Bonaccorso et al. 2003; Raziei et al. 2008; Santos et al. 2010; Vicente-Serrano 2006).

The capability of each of these drought-related factors including precipitation, vegetation, and land surface temperature in drought monitoring varies along seasonal time. Vegetation shows different sensitivities to water stress during different phenological phases. Thus, appropriate weights should be assigned to these factors in principal component analysis at different time periods. Based on the PCA method, this study aims to establish a new remote sensing index named Integrated Drought Condition Index (IDCI), which considers precipitation, vegetation growth condition, and land surface temperature comprehensively. In this index, these factors are weighed differently for each month of the growing season.

2 Study area and data

2.1 Study area

The study area is situated in Northern China with a latitude of $33^{\circ}54'–46^{\circ}43'N$ and longitude of $107^{\circ}54'–131^{\circ}41'E$ (Fig. 1). The total area is $163.69 \times 10^4 \text{ km}^2$ and covers 14 provinces, including Hebei, Shandong, Shanxi, and Liaoning. The major land cover categories of the region are grasslands, croplands, and mixed forest according to the 2012 MODIS land cover classifications data (MCD12Q1). In addition, this region is mainly located in the arid/semiarid area of the mid-temperate zone and in the humid/sub-humid area of the warm temperate zone based on the Chinese eco-geographical zoning map (Zheng and Li 2008). The East Asian monsoon climate produces uneven precipitation spatial distribution. Northern China suffers from frequent climate-related disasters, such as droughts and floods.

2.2 Data

2.2.1 *In situ meteorological data*

The monthly precipitation and mean temperature data from 1961 to 2012 of all the available stations were acquired from China Meteorological Data Sharing Service System

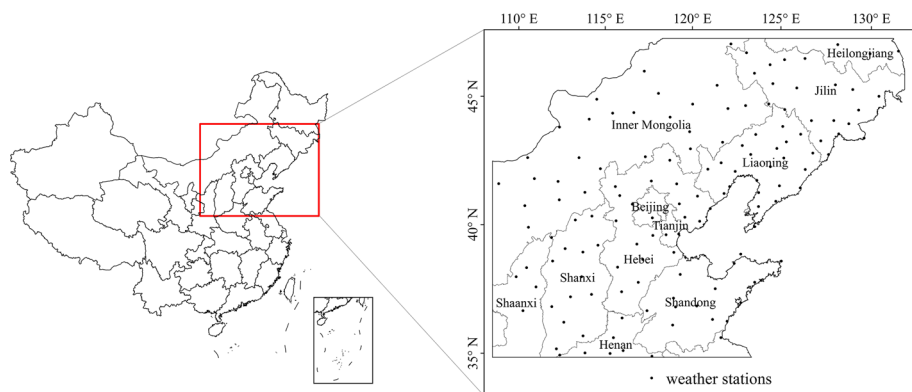


Fig. 1 Study area and the distribution of the meteorological stations

(<http://cdc.cma.gov.cn/>). The meteorological data records were inspected rigorously. Only those meteorological stations with available long-term data from 1961 to 2012 were included. A total of 135 meteorological stations in the study area were selected (Fig. 1). The monthly precipitation and mean temperature for all the meteorological stations within each province were averaged to represent the regional drought condition.

The Standardized Precipitation Evapotranspiration Index (SPEI) developed by Vicente-Serrano et al. (2010a) can be flexibly designed to measure drought severity for a specific time period based on the long-term precipitation and PET record. In this study, PET was calculated based on the Thornthwaite equation (Thornthwaite 1948) which considers only monthly mean temperature data. Different timescales are useful for monitoring different drought types. Short SPEI timescales are mainly appropriate for analyzing soil water content and river discharge in headwater areas, while long SPEI timescales are related to variations in groundwater storage (Vicente-Serrano et al. 2010a). Given that the major interest of this study is short-term drought, the multi-scale SPEIs (1, 3, and 6 months) of 135 weather stations were calculated using the long-term monthly precipitation and mean temperature data from 1961 to 2012.

The percentage of precipitation anomalies (Pa) is a simple indicator in measuring precipitation anomalies compared with the historical average for a specific period (Zhou et al. 2012). In this study, we calculated the Pa using the monthly precipitation data for the weather stations throughout the study area.

2.2.2 Remote sensing data

The Terra MODIS 8-day average value of composited LST products with 1 km resolution (MOD11A2, collection v005) from 2003 to 2012 and the monthly maximum value composited (MVC) MODIS NDVI time series products with 1 km resolution (MOD13A3, collection v005) from 2003 to 2012 in four tiles (h26v04, h26v05, h27v04, and h27v05) that covered the study area were obtained from the National Aeronautics and Space Administration (NASA) Land Processes Distributed Active Archive Center. The original NDVI and LST images were spliced and their projection system was converted from sinusoidal to an Albers conical equal area projection using the MODIS reprojection tool developed by NASA.

The remotely sensed precipitation data used in this study were derived from the TRMM satellite that was launched in November 1997. The TRMM 3B43 dataset from 2003 to 2012 was obtained from the NASA Data and Information Services Center (<http://mirador.gsfc.nasa.gov/>). This dataset was presented as a monthly precipitation rate (mm/h) in a $0.25^\circ \times 0.25^\circ$ spatial resolution and covered the 50° south to 50° north global latitude band. To explore the optimal integrated drought index, accumulated 3- and 6-month TRMM data were calculated and the 1-, 3-, and 6-month TRMM data were tested, respectively, in the subsequent experiment.

3 Methodology

3.1 Remote sensing data processing

The NDVI and LST datasets were MODIS monthly MVC and 8-day average value products, respectively. However, noises still existed because of cloud contamination and

atmospheric variability. These cloud-contaminated pixels in the LST and NDVI products were masked out based on the quality control documents. Furthermore, to maintain a temporal resolution that is the same as that of the NDVI dataset, the 8-day LST was composited to monthly LST by calculating the average value weighting with the number of days belonging to each month after masking the fill and missing values (Rhee et al. 2010).

All the remote sensing variable values, including the NDVI, LST, and 1- to 6-month accumulated TRMM precipitation data, were linearly scaled from 0 to 1 for each pixel based on the absolute maximum and minimum values of the same month in the entire record (2003–2012) to discriminate the weather and ecosystem components. The process adhered to that performed by Kogan (1995) for the VCI using the NDVI. Moreover, the scaled value was changed from 0 to 1 to correspond to the driest and wettest conditions. In addition, the scaled TRMM precipitation data were resampled to 1 km resolution using a bilinear interpolation method to ensure that spatial resolution is consistent with that of MODIS data. The formulas for each remote sensing drought indices are given in Table 1.

3.2 Remote sensing drought-related factors and the principal component analysis method

Since the relationship between different remotely sensed drought indices and SPEI varies over time, Pearson's correlation analyses were performed on the remote sensing drought indices of each month from May to September and SPEI of different timescales from 2003 to 2012. These analyses were conducted to assess the capability of these drought-related factors in drought monitoring and further determine the appropriate weights. The remotely sensed index values at the weather station were extracted using the $3 \text{ km} \times 3 \text{ km}$ window method. The information from the $3 \text{ km} \times 3 \text{ km}$ pixel window that centered on each weather station location was extracted, and the average value of all pixels in the extraction window was calculated to represent the index value of the site location.

The PCA method, a powerful tool for data analysis, was used to obtain the integrated information from the TCI, VCI, and scaled TRMM. The capability of different indices in monitoring drought showed considerable variations along seasonal time; thus, we weighted these indices differently over different time periods while performing PCA. The integrated drought condition indices with different weights were tested through Pearson's correlation analyses.

3.3 Evaluation of IDCI drought monitoring behavior using in situ drought observations

An IDCI dataset that contains optimal weights during the growing season (May to September) from 2003 to 2012 was created based on the PCA method. The spatial changes in the IDCI maps were compared with the changes in the in situ drought indices in a typical

Table 1 Remote sensing drought index formulas

Drought index	Formula
Scaled LST (modified TCI)	$(LST_{\max} - LST)/(LST_{\max} - LST_{\min})$
Scaled NDVI (VCI)	$(NDVI - NDVI_{\min})/(NDVI_{\max} - NDVI_{\min})$
Scaled TRMM	$(TRMM - TRMM_{\min})/(TRMM_{\max} - TRMM_{\min})$

The scaled TRMM uses one of the 1-, 3-, and 6-month timescales

dry year and individual month of August from 2003 to 2012 to assess the IDCI performance in monitoring short-term drought conditions. The spatial distributions of the areas that experienced drought were compared.

Pearson's correlation analyses were performed between the IDCI values and the 1-, 3-, and 6-month SPEI values during the growing season from 2003 to 2012. Additionally, regression analyses were performed between the IDCI and the in situ reference data from 2003 to 2012 to further evaluate the capability of regional drought condition monitoring. Analyses were performed in nine main provinces of the study area. Furthermore, the yearly IDCI variations were compared with the yearly variations in the in situ drought indices to evaluate the temporal drought monitoring capability of this index. The IDCI values from 2003 to 2012 at the weather stations were the average value of all pixels in the $3 \text{ km} \times 3 \text{ km}$ window which centered on the in situ site location.

4 Results and discussion

4.1 Correlations between drought-related factors and SPEIs

The monthly correlation coefficients were calculated between the remotely sensed drought indices (TCI, VCI, and scaled TRMM) and the 1-, 3-, and 6-month SPEI (SPEIs) from 2003 to 2012 (Fig. 2). The correlation coefficient values varied among the different timescales and were statistically significant at the 0.01 significance level except for the VCI versus SPEI-1 in May and September (Fig. 2a). Notably, the scaled TRMM with various timescales showed high correlation coefficient values with the in situ drought index, followed by the TCI and VCI (Fig. 2). Overall, the scaled TRMM mostly produced the highest correlation coefficient values with in situ variables with the same timescales as that of the scaled TRMM. For example, the scaled TRMM1 (1-month TRMM data used) was best correlated with SPEI-1 (Fig. 2a), the scaled TRMM6 (6-month TRMM data used) was best correlated with SPEI-6 (Fig. 2c), and the scaled TRMM3 (3-month TRMM data used) was best correlated with SPEI-3 except in May (Fig. 2b).

In addition, compared with that of the scaled TRMM and TCI, the correlations between the VCI and the SPEIs varied significantly among the different SPEI timescales and over the different time periods. The VCI appeared less sensitive to SPEI-1 than to SPEI-3 or SPEI-6 (Fig. 2a) because there is a time lag existed between the precipitation occurrence and the vegetation condition response. The VCI showed the highest correlation with the SPEIs in July because of the vegetation phenological phase in the study area (Fig. 2).

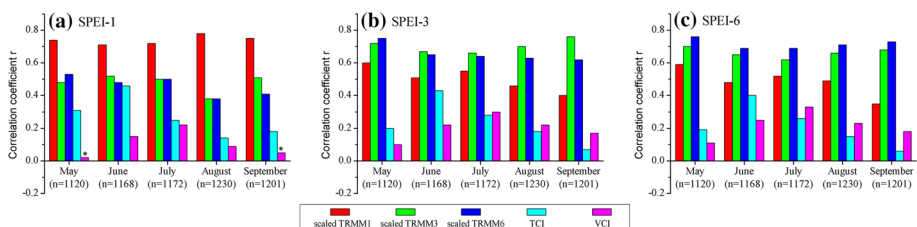


Fig. 2 Correlations between the remote sensing drought indices and the in situ different timescales SPEI from May to September. The p value < 0.01 except for columns with an asterisk (*). The descriptions of the drought indices are given in Table 1. SPEI-1, 1-month SPEI; SPEI-3, 3-month SPEI; SPEI-6, 6-month SPEI

Table 2 Correlation coefficient r values between the integrated drought condition indices with different weights and the in situ different timescales SPEI from May to September

Index	Month	Weight			r		
		Scaled TRMM (α)	TCI (β)	VCI (γ)	SPEI-1 (TRMM1 used)	SPEI-3 (TRMM3 used)	SPEI-6 (TRMM6 used)
IDCI	a. May ($n = 1120$)	1	1	1	0.14	0.25	0.25
		2	0	1	0.20	0.27	0.26
		2	1	0	0.64	0.59	0.62
		3	1	1	0.30	0.30	0.28
		4	0	1	0.45	0.33	0.29
		4	1	0	0.64	0.59	0.63
		4	3	1	0.45	0.33	0.29
		5	3	2	0.25	0.29	0.27
	b. June ($n = 1168$)	1	1	1	0.28	0.24	0.28
		2	0	1	0.41	0.30	0.30
		2	1	0	0.56	0.60	0.65
		3	2	0	0.57	0.60	0.65
		3	2	2	0.37	0.27	0.29
		4	1	0	0.57	0.59	0.64
		4	3	1	0.57	0.45	0.43
		5	2	1	0.57	0.48	0.47
	c. July ($n = 1172$)	1	1	1	0.55	0.44	0.42
		2	1	1	0.64	0.52	0.53
		3	1	2	0.61	0.47	0.49
		3	2	2	0.61	0.48	0.49
		4	1	3	0.59	0.47	0.47
		4	3	1	0.64	0.57	0.61
		5	1	2	0.64	0.55	0.56
		5	2	1	0.64	0.57	0.62
	d. August ($n = 1230$)	1	1	1	0.66	0.51	0.49
		2	1	1	0.75	0.61	0.62
		3	0	2	0.74	0.58	0.58
		3	2	2	0.74	0.59	0.58
		4	0	3	0.73	0.58	0.55
		4	1	1	0.75	0.62	0.65
		4	1	3	0.73	0.58	0.55
		5	3	2	0.75	0.62	0.64
	e. September ($n = 1201$)	1	1	1	0.53	0.47	0.52
		2	0	1	0.60	0.68	0.64
		2	1	1	0.60	0.68	0.63
		3	0	2	0.58	0.64	0.62
		3	1	2	0.58	0.64	0.61
		4	0	1	0.67	0.68	0.64

Table 2 continued

Index	Month	Weight			<i>r</i>		
		Scaled TRMM (α)	TCI (β)	VCI (γ)	SPEI-1 (TRMM1 used)	SPEI-3 (TRMM3 used)	SPEI-6 (TRMM6 used)
	4		1	3	0.57	0.63	0.60
	5		1	2	0.65	0.68	0.64

In all the cases, the p value < 0.01 . The highest r values for each column at different time periods are shown in bold. The descriptions of the drought indices are presented in Table 1. SPEI-1, 1-month SPEI; SPEI-3, 3-month SPEI; SPEI-6, 6-month SPEI

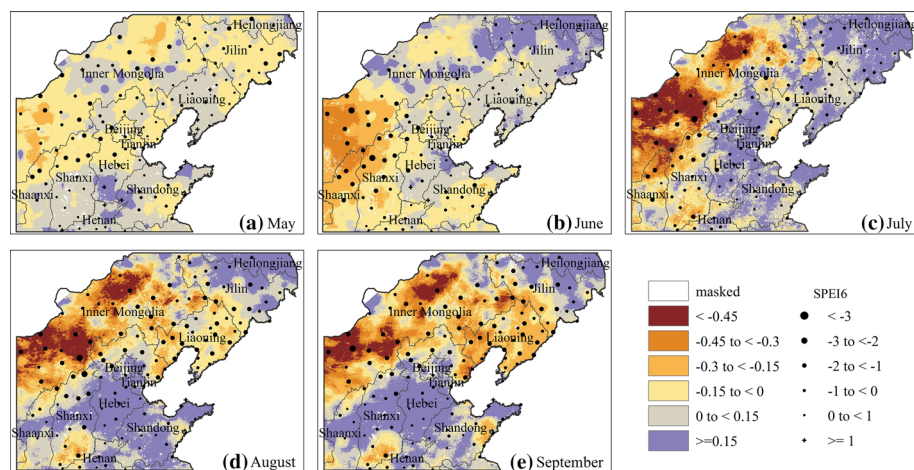


Fig. 3 Seasonal changes in drought detected by IDCI (using TRMM6) and SPEI-6 from May to September in 2009. SPEI-6, 6-month SPEI

Additionally, compared with the VCI, the TCI demonstrated more correlation with both 3- and 6-month SPEI from May to June. This result indicates that the TCI can provide valuable drought information during that time period (Fig. 2b, c).

Generally, the scaled TRMM showed a higher correlation with the in situ drought indices than the TCI and VCI, and the capability of the TCI and VCI in drought monitoring highly differed over different time periods. Therefore, appropriate weights were assigned to these indices in the PCA at different time periods. The original spectral feature space in the PCA procedure was composed of the scaled TRMM, TCI, and VCI, as shown in the following:

$$\underbrace{[\text{scaled TRMM}, \dots, \text{scaled TRMM}]_{\alpha}}_{\alpha} \underbrace{[\text{TCI}, \dots, \text{TCI}]_{\beta}}_{\beta} \underbrace{[\text{VCI}, \dots, \text{VCI}]_{\gamma}}_{\gamma}$$

where α , β , and γ show the number of occurrences of these three indices in the original feature matrix and represent the weights of these three indices, respectively. The scaled TRMM used one of the 1-, 3-, and 6-month timescales. The covariance matrix and the

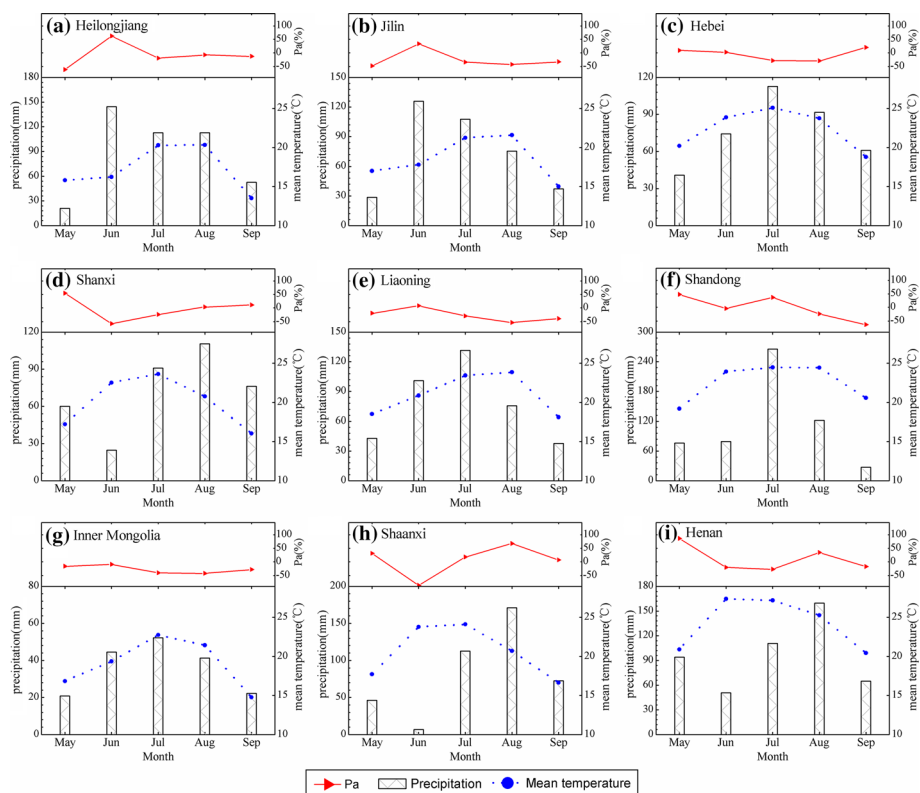
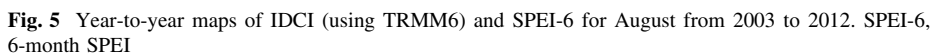


Fig. 4 Seasonal variations of precipitation, mean temperature, and the percentage of precipitation anomalies (Pa) in nine provinces from May to September in 2009

eigenvalues and eigenvectors of the covariance matrix were calculated. Subsequently, a series of new principal components was computed by multiplying the eigenvector of the covariance matrix for the original feature matrix. The Integrated Drought Condition Index was defined as the first principal component, by considering that the first principal component contained the most number of information from the original spectral feature space.

To select the optimal IDCI weight over different time periods, Pearson's correlation analyses were performed between the integrated drought condition indices and the 1-, 3-, and 6-month SPEI values for each month of the growing season (Table 2). All the correlations were statistically significant at the 0.01 significance level. The r values generally decreased as the VCI weight increased in May and June (Table 2a, b). By contrast, the integrated drought condition indices with larger VCI weight yielded higher r values with the in situ drought indices from July to September than that in May and June (Table 2c–e). Moreover, during the entire growing season, the integrated drought condition indices with appropriate weights were better correlated with the in situ variables than the indices with equal weights of 1, 1, and 1 for the scaled TRMM, TCI, and VCI, respectively (Table 2). Based on the relatively higher correlations between the IDCI with in situ drought indices over different time periods, the May to September optimal weights (α , β , γ) for the scaled TRMM, TCI, and VCI of the IDCI were (4, 1, 0), (3, 2, 0), (5, 2, 1), (4, 1, 1), and (4, 0, 1), respectively (Table 2). Finally, an IDCI dataset with different weights was obtained during



4.2 Drought conditions monitored by IDCI

Although local discrepancies between SPEI-6 and IDCI values were found, the two variables basically showed a similar spatial pattern. Based on the IDCI maps, the drought

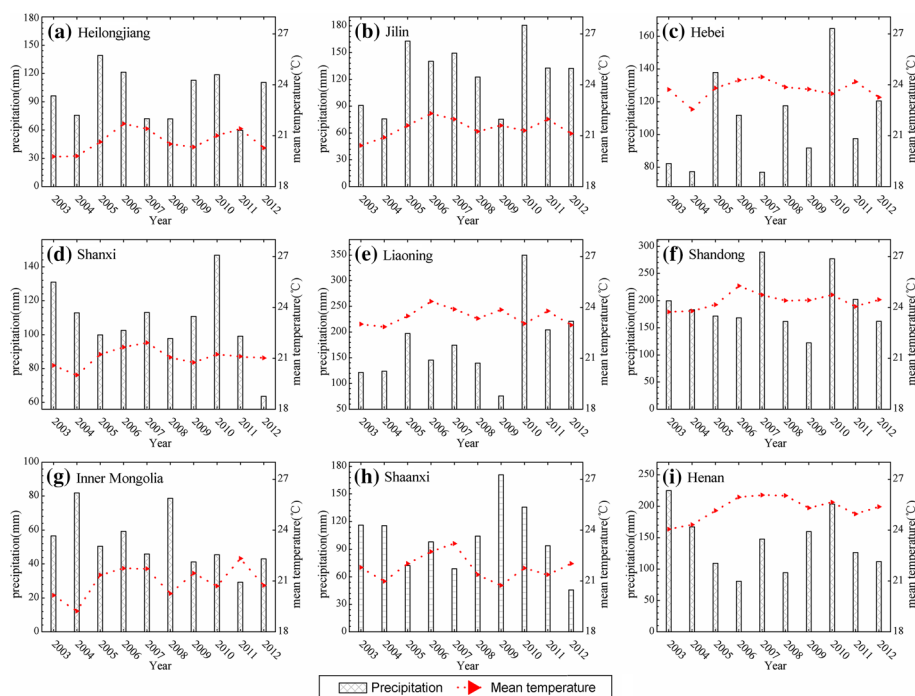


Fig. 6 Annual variation of the precipitation and mean temperature data in nine provinces for August from 2003 to 2012. SPEI-6, 6-month SPEI

in Northeast part (Jilin and Liaoning) developed heavily from July to September 2009 (Fig. 3c–e). These situations are evidenced by the SPEI-6 values in those regions, which tend to be lower (Fig. 3c–e), indicating that precipitation deficiency and high temperature were affecting those areas (Fig. 4b, e). In June 2009, IDCI showed more severe drought conditions in northern Shaanxi because the average precipitation of the region was only 6.9 mm and the Pa value was -86% (Figs. 3b, 4h). SPEI-6 showed a similar trend with lower values (Fig. 3b). Inner Mongolia was affected by serious drought conditions (Fig. 5c, g, i), because of the precipitation declining and high temperature in August of 2005, 2009, and 2011 (Fig. 6g). SPEI-6 values in the corresponding time were low, which showed agreement with the IDCI pattern (Fig. 5c, g, i). Moreover, the SPEI-6 showed more severe conditions in Jilin in August 2004 as well as in northern Henan and Shandong in August 2012. The same severity and extent of the droughts were also detected through the IDCI maps (Figs. 5b, j).

4.3 Correlation analysis and year-to-year comparisons

The IDCI values were compared with the 1-, 3-, and 6-month SPEI values for each month of the growing season from 2003 to 2012 (Table 3). The results showed that the IDCI and SPEIs were highly correlated during the growing season, and all the correlations were statistically significant ($p < 0.005$). The IDCI values for the nine main provinces (Heilongjiang, Jilin, Hebei, Shanxi, Liaoning, Shandong, Inner Mongolia, Shaanxi, and Henan) were also tested. The scatter plots between the monthly IDCI (using TRMM6) and the SPEI-6 for each province with weather stations are presented in Fig. 7. The IDCI agreed

Table 3 Correlation coefficient r values between the IDCI and the in situ different timescales SPEI during the growing season from 2003 to 2012

Index	Year	May		June		July		August		September							
		SPEI-1	SPEI-3	SPEI-6	SPEI-1	SPEI-3	SPEI-6	SPEI-1	SPEI-3	SPEI-6	SPEI-1	SPEI-3	SPEI-6				
<i>r</i>	IDCI	2003	0.75	0.82	0.90	0.57	0.61	0.74	0.65	0.59	0.68	0.76	0.65	0.68	0.72	0.75	0.75
	2004	0.73	0.76	0.82	0.55	0.66	0.72	0.69	0.60	0.66	0.77	0.60	0.60	0.58	0.63	0.63	0.61
	2005	0.76	0.82	0.84	0.75	0.83	0.85	0.69	0.79	0.83	0.71	0.79	0.79	0.84	0.85	0.78	0.82
	2006	0.76	0.55	0.49	0.65	0.67	0.68	0.62	0.56	0.60	0.72	0.46	0.46	0.54	0.74	0.53	0.50
	2007	0.60	0.68	0.66	0.71	0.65	0.68	0.60	0.60	0.60	0.56	0.83	0.70	0.70	0.79	0.75	0.76
	2008	0.68	0.51	0.55	0.56	0.39	0.49	0.69	0.40	0.41	0.63	0.40	0.40	0.39	0.79	0.54	0.37
	2009	0.75	0.63	0.65	0.78	0.65	0.70	0.60	0.60	0.47	0.61	0.82	0.62	0.56	0.61	0.77	0.61
	2010	0.67	0.70	0.74	0.74	0.43	0.40	0.46	0.86	0.70	0.78	0.84	0.81	0.81	0.69	0.81	0.75
	2011	0.55	0.48	0.63	0.63	0.63	0.55	0.54	0.68	0.57	0.60	0.73	0.59	0.57	0.74	0.75	0.65
	2012	0.77	0.56	0.53	0.79	0.81	0.82	0.82	0.63	0.79	0.77	0.81	0.69	0.69	0.74	0.67	0.76

In all the cases, the p value <0.005 . SPEI-1, 1-month SPEI; SPEI-3, 3-month SPEI; SPEI-6, 6-month SPEI

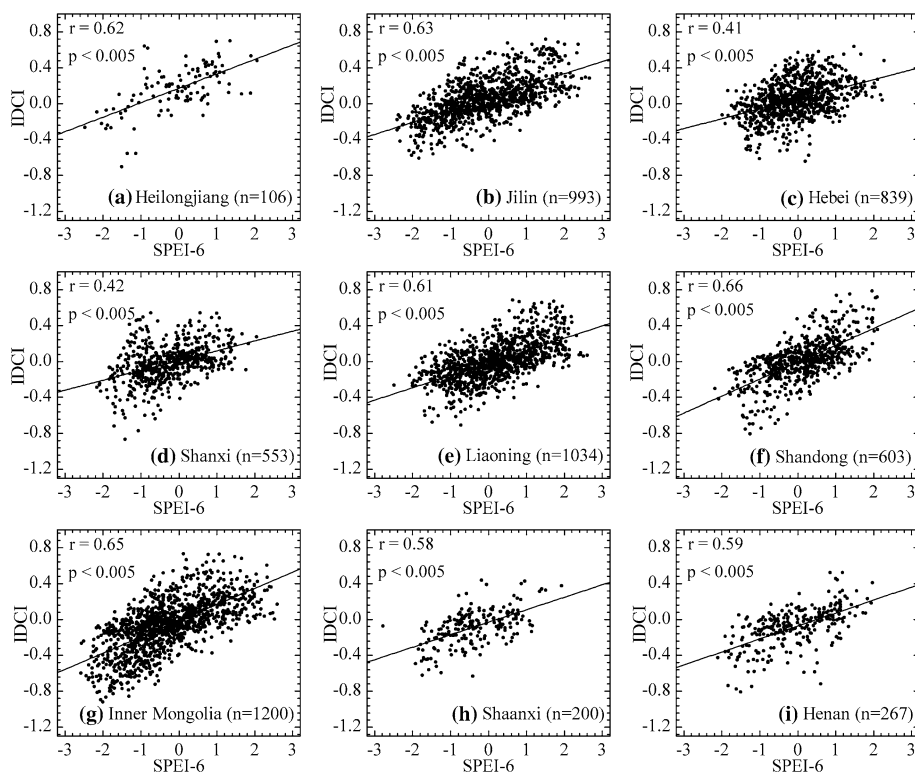


Fig. 7 Scatter plots and correlation coefficient r values between the IDCI (using TRMM6) and SPEI-6 in nine provinces. SPEI-6, 6-month SPEI

well with the SPEI-6 in different provinces during the growing season (Fig. 7), and all the correlations passed the p value < 0.005 significance test.

The yearly IDCI changes (using TRMM-6) in 18 representative meteorological sites were compared with that of the SPEI-6 for August (Fig. 8). These 18 representative meteorological sites, which are evenly distributed, can represent the three main land cover types (grasslands, croplands, and mixed forest) in the study region. There are some discrepancies between the IDCI and the SPEI-6. For example, the SPEI-6 slightly decreased from 2009 to 2010, whereas the IDCI increased in Yan'an (Fig. 8h). The SPEI-6 increased from 2007 to 2008 in Yiyuan, whereas the change in the IDCI had the opposite result (Fig. 8r). In most cases, the IDCI exhibited consistent variations with the in situ reference data at the regional scales, indicating that IDCI can be used to characterize drought conditions and patterns effectively.

5 Conclusion

This study established a new remote sensing index named Integrated Drought Condition Index (IDCI) using the PCA method. This synthesized drought index was found to be capable of integrating precipitation, vegetation growth condition, and land surface temperature with the monthly optimal weight during the growing season.

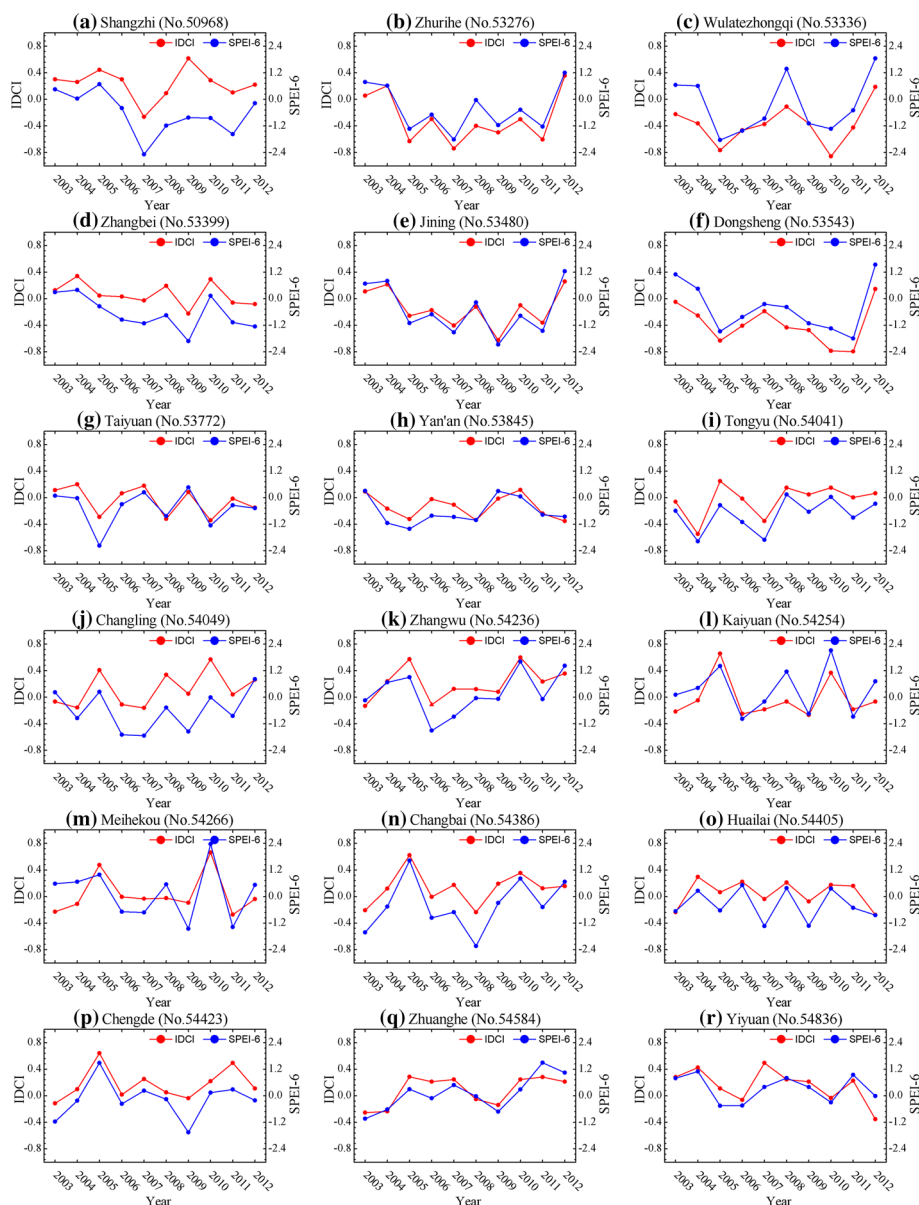


Fig. 8 Year-to-year changes in the IDCI (using TRMM6) and SPEI-6 in August from 2003 to 2012. SPEI-6, 6-month SPEI

The regional drought monitoring capability of the IDCI was validated by comparing the seasonal changes and inter-annual variations in the IDCI maps and in the in situ reference data. The IDCI was found to be effective in characterizing drought conditions and patterns. The IDCI exhibited high correlations with the in situ SPEIs. The yearly IDCI variations agreed well with the in situ reference data variations. However, the efficiency of IDCI may

be improved by taking into consideration the soil moisture factor because it is an important indicator of drought events. Therefore, in future research, the soil moisture data will become a component of the Integrated Drought Condition Index to provide more detailed drought condition information.

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